BYTEENGINE
Systems you can reuse

Source audit, independent authoring guide and Headwater Level 5 reference
19 October 2026

Collection	Audit result
Dune Company	Plugin source and installed DLLs match repository packages. Native component classes and Add Component entries verified.
Headwater	Native components, editable Event Sheets and blueprints. No gameplay plugin required. Native classes and Add Component entries verified.
Downhill / Legends Road	Missing canonical source recovered before you deleted the project. Its installed package matched the repository package at that point. The deleted project stays deleted.

Fixed: Downhill source is preserved under Plugins/Downhill. Tools/AuditProjectSystems.ps1 provides a repeatable check. Mechanical Latch, Primed Siphon and Water Counterweight Bridge are now native engine components.

Meaning of a match: current project plugin source files match canonical source. Installed plugin IDs, versions and DLL hashes match the repository library. Native class definitions and editor Add Component registrations exist. These checks do not prove every possible game configuration works.

Scope: Dune Company and Headwater are the remaining live projects. Historical Karlson, Karlson Demo, Goblin Scrapper and Space Scraper plugin collections remain available. Deleted projects were not treated as live tested projects. Local repository changes do not imply a GitHub push.

02 / Make another game yourself

1. Create the game in its own folder under Documents using the editor project browser. Keep game scenes and assets outside the engine repository.

2. Select an object, use Add Component and search for the native system by name. Edit references and tuning in the Inspector. Combine small components rather than copying gameplay code.

3. For a plugin, open Project Settings, select Plugins and choose Import Plugin. Browse to a .byteplugin package in C:/Users/codex/ByteEngine/PluginLibrary. Reopen the project to load its registrations.

4. Add the plugin components through Add Component. Public tuning properties appear in the Inspector. Event Sheets can read state properties and change writable component properties.

5. Save repeated object hierarchies as blueprints, keeping their linked Event Sheet assets. A blueprint captures structure and settings; the Event Sheet supplies the level rules.

6. Test a small scene first: start, action, pause, retry and completion. Confirm platform support before building a full game.

Location	Purpose
Engine/ByteEngine.Core	Reusable native behavior, serialization and engine systems.
Plugins/<name>	Canonical C# source for a domain or game plugin.
PluginLibrary	Installable packages for Project Settings / Plugins.
Documents/<game>/Assets	Models, audio, editable Event Sheets and blueprints.
Documents/<game>/Scenes	Level layouts and component settings.

The project Tools folders contain authoring and validation helpers. Playing and editing the resulting scenes does not require running those helpers.

03 / Native hydraulic building blocks

Add Component name	Setup and responsibility
Water Network	One coordinator finds basins, gates, streams and buoyant bodies at scene start. Conserves water volume.
Water Basin	Bounds, Floor, Rim and Initial Level define a finite control volume. Reference surface and volume visual objects by name.
Water Gate / Pump	Source and Destination names, Conductance, Opening, One Way, Overflow and Crest Height. Is Pump explicitly enables uphill transfer.
Screw Sluice Actuator	A scene boolean drives finite blade travel. Set Closed Height, Travel, Speed and linked Wheel Name. Water Gates read measured aperture.
Buoyant Body + Rigidbody	Displaced water and current drag act on a real rigid body. Set collision, dimensions and sensible mass.
Buoyant Pulley	Float Name, Load Name, Rest Height, Ratio and Maximum Travel. Real float displacement raises the load.
Water Current Path / Stream Transport	Currents use measured delivered flow. Streams store water in transit; connect the gate Transit Path Name.
Cargo Dock Receiver	Carrier and Cargo, capture volume, mooring and unload destination. Both bodies must enter the zone.
Timed Challenge	Start and Finish variables, a unique stable Record Key and star thresholds. Best saved results drive level selection.

Small tutorial: create two Water Basins, a Water Gate and one Water Network. Give the higher basin water. Set the gate Source and Destination to their object names and Opening to zero. Use a pointer target and Event Sheet to switch Opening between zero and one. Add a visible stream after the conserved transfer works.

Keep switches in scene variables and consequences tied to actual level, travel or delivered flow. Avoid guessing mechanism state from a timer.

04 / The three new native systems

Mechanical Latch

Add it to a compatible load object. Command Variable requests pin insertion; Can Engage is eligibility supplied by events. Pin Name and Pin Travel move a separate pin model. Read Latched and Engagement for feedback. Buoyant Pulley and Water Counterweight Bridge respect Latched, so draining supporting water does not fight a second transform writer.

Try it: attach a latch to the pulley gate. Set its command to GatePinned. In events set Can Engage when pulley Travel reaches the raised threshold. A pointer toggles GatePinned. Once Latched is true, drain the float and verify the blade holds.

Primed Siphon

Set Source Basin, Flow Gate, Prime Height and Air Break Height. Rising source water starts the connected conserved gate. Flow continues below the prime line until air reaches the intake. It creates no water and is not an uphill pump. Read Primed, Prime Count and Source Level in events.

Try it: place the source above its receiver and below its prime line initially. Add supply, then close supply after priming. Check that siphoning continues to the air break. Give the receiver enough capacity or a real drain.

Water Counterweight Bridge

Add it at the deck hinge and set Ballast Basin. Dry Mass, Counterweight Mass and Kilograms Per Degree map finite water volume to a target angle. Response and Maximum Degrees Per Second smooth motion. Rotation Axis must match the hinge. Alignment Angle and Tolerance define Is Aligned. A Mechanical Latch on the same object holds the angle after draining ballast.

Try it: fill an empty ballast basin, inspect Angle, then trim water through a drain. Pin inside the alignment band. Test real collider clearance for both raft and cargo.

Canonical source: Engine/ByteEngine.Core/Gameplay/Hydraulics/WaterPuzzleMechanisms3D.cs. Save/load registration is in ComponentSerializer.cs; Add Component metadata is in ComponentMetadataRegistry.cs. Headwater-specific rules remain in project events.

05 / Plugins and platform limits

Package	Version / purpose
DesertTerrain	0.8.11 - procedural terrain used by Dune Company.
DuneCompany	0.27.0 - workshop, vehicles and Dune game rules. Domain-specific, not a universal game controller.
Downhill	0.4.0 - board controller, follow camera, rider, feedback and run rules. Recovered source is under Plugins/Downhill.
Karlson / KarlsonDemo	1.0.1 / 2.4.0 - original FPS and modular weapon gameplay collections.
GoblinScrapper / SpaceScraper	0.4.5 / 0.1.0 - retained construction and gameplay plugin collections.

Desktop: import a packaged plugin, reopen and configure its components. A matching package does not automatically make every plugin portable to every platform.

Browser: native engine components compile into the browser runtime. The current linked plugin set includes DesertTerrain and DuneCompany. Downhill and the other historical plugins are not currently advertised as browser-ready. Installing their package alone does not link executable plugin code into the browser runtime.

Editing plugin C#: edit canonical Plugins source, increment the version, build its .csproj and use the package under artifacts/plugins. Synchronize project Plugins, project PluginLibrary, engine PluginLibrary and project PluginSource. Keep the same version throughout.

Dune-specific rule: after changing Dune plugin code, refresh the statically linked browser runtime with Tools/PublishBrowser.ps1 and synchronize Dist and Debug copies. Runtime publishing is separate from game export. No Dune game export was requested here.

Repeat the check: run Tools/AuditProjectSystems.ps1 in PowerShell from the engine folder. Defaults cover Dune Company and Headwater. Pass -Projects to include other folders; -OutputPath writes JSON. A nonzero exit indicates a missing project or mismatch.

The native checks inspect scene and blueprint types for engine source and Add Component entries. They do not validate every property, asset reference, Event Sheet rule or browser interaction. Use gameplay tests alongside the source audit.

06 / Headwater / The Tideworks

Open C:/Users/codex/Documents/Headwater/Headwater.byteproject. Level 5 is Scenes/Level05.bytescene. Completing Level 4 unlocks its card. For testing, open Select Level and type HEADWATER. Test access unlocks cards without awarding stars; End Test Access restores progression.

Player sequence

1. Fill the float and insert the raised gate pin.

2. Drain its supporting water to carry cargo under the low roof.

3. Route the distribution fork to the cistern. Fill to the priming line; the siphon fills the bridge ballast.

4. Trim ballast until the bridge enters the 54-62 degree pin range. Pin it, open the berth and keep recovery water moving.

5. Wait for the WHOLE raft to enter the final lock. Close the entrance and turn on the recovery pump. It transfers existing canal water to lift cargo to the dock.

Editable assets

Blueprint: Assets/Blueprints/Level 5 - Tideworks machinery.byteblueprint.
Rules: Assets/Events/Level 5 - The Tideworks.byteevents.
Select a basin, gate, linkage, bridge or pointer target to edit its native settings. Events control distribution, pin eligibility, berth interlock, lock balance, hints and delivery.

Validation

The native scripted solution completed in 54.4 simulated seconds. Checks cover idle stability, controls, pause, latch eligibility, priming, bridge pinning, raft entry and physical unload. Maximum water-volume error: 3.98e-13; maximum tracked raft movement per frame: 0.051 m. These are solution checks, not human difficulty testing or an FPS guarantee.

Separate engine regression tests cover siphon hysteresis, air break, conserved transfer and bridge hold/release. Menu tests cover fresh locks, saved stars, test access, ending test access and loading the fifth card.

Level 5 music: Water Prelude by Kevin MacLeod, CC BY 4.0. Edited into a normalized crossfaded loop. Full attribution is in game credits and Assets/Audio/CREDITS.md. Official track: incompetech.com/music/royalty-free/index.html?Search=Search&isrc=USUAN1100017